Abstract

Recent work on time-series anomaly detection reports high F1 scores on standard benchmarks, but many of these results rely on point adjustment, which can inflate performance by treating an entire anomaly segment as detected once any single point in the segment is flagged. In this work, we re-evaluate time-series anomaly detection methods using raw point-level PR-AUC, computed directly from anomaly scores without point adjustment. We compare PWTA with five classical baselines and AnomalyTransformer, and we analyze the score degeneracy of AnomalyTransformer and DCdetector.

Our results show that AnomalyTransformer reproduces its reported point-adjusted F1 scores but performs close to the random baseline under raw PR-AUC. A similar score degeneracy is observed for DCdetector on SWaT, where most anomaly scores are exactly zero and the remaining non-zero scores provide little separation between normal and anomalous regions. In contrast, PWTA shows stronger raw point-level discrimination than the classical baselines on the multi-entity benchmarks, ranking first on 51 of 110 entities. PWTA also outperforms AnomalyTransformer under raw PR-AUC on SMD, SMAP, and MSL.

The advantage of PWTA has a clear boundary. It performs well on datasets where anomalies appear as temporal or cross-channel pattern changes, but it does not dominate on SWaT, where many anomalies remain within the normal value range and Isolation Forest performs better. These findings suggest that the main contribution of this work is not a universally superior architecture, but a corrected evaluation protocol and an empirical analysis of which types of anomaly detectors remain effective under that protocol.

Keywords time-series anomaly detection, point adjustment, PR-AUC, Transformer autoencoder, reconstruction

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Chapter 1. Introduction

Time-series data are continuously produced by many real-world systems, including server infrastructure, spacecraft telemetry, and industrial control processes. In these systems, abnormal patterns may indicate service failures, equipment faults, cyber-physical attacks, or operational risks. Detecting such patterns early is therefore important for reducing downtime, preventing safety issues, and supporting timely maintenance. Time-series anomaly detection (TSAD) addresses this problem by identifying deviations from normal behavior in sequential observations.

Despite its practical importance, TSAD remains difficult for several reasons. Anomalies are rare, often accounting for only a small fraction of all time points, and labeled anomalies are expensive to obtain. In addition, the meaning of an anomaly varies across domains. Some anomalies appear as clear changes in value, such as sudden spikes or drops, while others remain within the normal value range and are visible only through temporal or cross-channel patterns. This diversity makes both detection and evaluation challenging, because a detector that works well for one type of anomaly may fail for another.

Recent TSAD research has increasingly used Transformer-based architectures. Xu et al. (2022) proposed AnomalyTransformer, which introduced an association-discrepancy criterion, and Yang et al. (2023) proposed DCdetector, which uses a dual-attention contrastive representation. These models report F1 scores above 0.9 on standard benchmarks such as SMD, MSL, SMAP, PSM, and SWaT. At first glance, these numbers suggest substantial progress. However, many reported results rely on point adjustment (PA), an evaluation convention that can strongly inflate F1 scores.

Under point adjustment, if a detector flags any single point inside a ground-truth anomaly segment, the entire segment is treated as correctly detected before computing F1. This rule may be reasonable when the goal is only to determine whether an event was noticed at least once, but it does not measure point-level anomaly-score discrimination. In long anomaly segments, even a weak or nearly random score sequence can receive high credit if it happens to hit one point in the segment. Therefore, high point-adjusted F1 does not necessarily imply that a model assigns consistently higher scores to anomalous time points than to normal ones.

In this work, we re-evaluate TSAD methods using raw precision-recall area under the curve (PR-AUC), computed directly from point-level anomaly scores without point adjustment. PR-AUC is appropriate for this setting because anomalies are rare and the positive class is highly imbalanced. We compare PWTA, a simple patch-wise Transformer autoencoder, with five classical baselines across 110 entities from SMD, SMAP, and MSL. We also compare PWTA with AnomalyTransformer under the same raw point-level metric and analyze the score degeneracy of AnomalyTransformer and DCdetector.

Our results show that AnomalyTransformer reproduces its reported point-adjusted F1 scores but performs near the random baseline under raw PR-AUC. A similar degeneracy is observed for DCdetector on SWaT, where most anomaly scores are exactly zero and the remaining non-zero scores provide little separation between normal and anomalous regions. In contrast, PWTA outperforms the classical baselines in the 110-entity comparison, ranking first on 51 entities, and achieves substantially higher raw PR-AUC than AnomalyTransformer on SMD, SMAP, and MSL.

Finally, we show that the advantage of PWTA has a clear boundary. PWTA performs well when anomalies disrupt temporal or cross-channel patterns, but it is less effective on SWaT, where many attacks remain within the normal value range and Isolation Forest is better

matched to the data. These results suggest that the main contribution of this work is not a universally superior architecture. Rather, the contribution is a corrected evaluation protocol and an empirical analysis showing which types of detectors remain effective when point adjustment is removed.

Chapter 2. Related Work

2.1 Criticism of point-adjust evaluation

Point adjustment (PA) was introduced to give credit for detecting an anomalous event once at least one point within the corresponding anomaly segment is detected (Xu et al., 2018). Figure 2 illustrates this rule: a single correct prediction inside a true anomaly segment is expanded to the entire segment before the metric is computed. Although this convention reflects the event-level nature of many TSAD applications, later work has shown that it can substantially distort performance estimates. Kim et al. (2022) demonstrated, both theoretically and empirically, that PA can make weak or even random anomaly scores appear competitive with state-of-the-art methods, and proposed PA%K as a stricter alternative. Doshi et al. (2022) further analyzed the mechanism by which PA rewards isolated hits within long anomaly segments, while Gim and Min (2023) examined modified adjustment rules designed to reduce this inflation. Paparrizos et al. (2022) approached the evaluation problem from a different perspective by proposing range-based measures, including VUS, that are less dependent on a single threshold. Together, these studies show that PA-based F1 is not a reliable measure of point-level anomaly discrimination. However, most of this literature focuses on exposing the weakness of the metric rather than identifying which methods remain effective when PA is removed.

2.2 Reconstruction-based detection

Another major line of TSAD research detects anomalies through reconstruction or forecasting error. These methods learn the structure of normal time-series data and assign high anomaly scores to points or windows that cannot be reconstructed or predicted well. Representative examples include Donut (Xu et al., 2018), the LSTM-based dynamic-thresholding method of Hundman et al. (2018), OmniAnomaly (Su et al., 2019), and USAD (Audibert et al., 2020). Reconstruction-based methods are attractive because they can be trained on normal data and can capture temporal regularities without explicit anomaly labels. At the same time, they are vulnerable to over-generalization: if the model reconstructs anomalous inputs too well, the reconstruction error no longer separates anomalies from normal behavior. PWTA belongs to this family of methods. This connection is important for our analysis, because the magnitude dependence of reconstruction error helps explain both the success and the failure cases discussed in Section 5.4.

2.3 Patch representations

Splitting a signal into patches and treating each as a token began in vision with ViT (Dosovitskiy et al., 2021) and reached time series with PatchTST (Nie et al., 2023). PatchTST is a forecasting model built on channel independence, which is the opposite of our reconstruction model: we deliberately mix channels inside each token. We borrow the patch representation only; the objective and the channel treatment differ (Section 3.2).

2.4 The gap we address

Prior studies have shown that point adjustment can produce misleadingly high F1 scores, especially when anomalies are evaluated as segments. These studies are important because they call into question a common evaluation practice in TSAD. However, most of them focus on the failure of the metric itself. They show that PA-based scores can overestimate performance, but they do not fully answer the next practical question: once PA is removed, which methods still show meaningful discrimination under a raw point-level metric?

This work addresses that question by evaluating classical baselines, Anomaly transformer, DCdetector and a simple patch-wise reconstruction model under the same raw PR-AUC protocol. The goal is not only to show that PA changes the ranking of methods, but also to identify what kind of model remains effective after the evaluation artifact is removed. We further examine the boundary of this effectiveness by distinguishing time-pattern anomalies from value-range anomalies, showing that the advantage of reconstruction-based detection depends on the type of abnormality present in the data.

Chapter 3. Method

3.1 Patch-wise Transformer Autoencoder (PWTA)

PWTA is a reconstruction-based detector trained on windows of normal time-series data. Given a multivariate window, the model divides it into non-overlapping temporal patches and treats each patch as a token. Unlike channel-independent patch models, each token contains all channels within the corresponding time interval, so the model can learn both temporal relations across patches and cross-channel structure within each patch. The sequence of patch tokens is projected into a latent space, combined with positional encodings, and passed through a Transformer encoder. A reconstruction head then maps the encoded tokens back to the original window space (Figure 1).

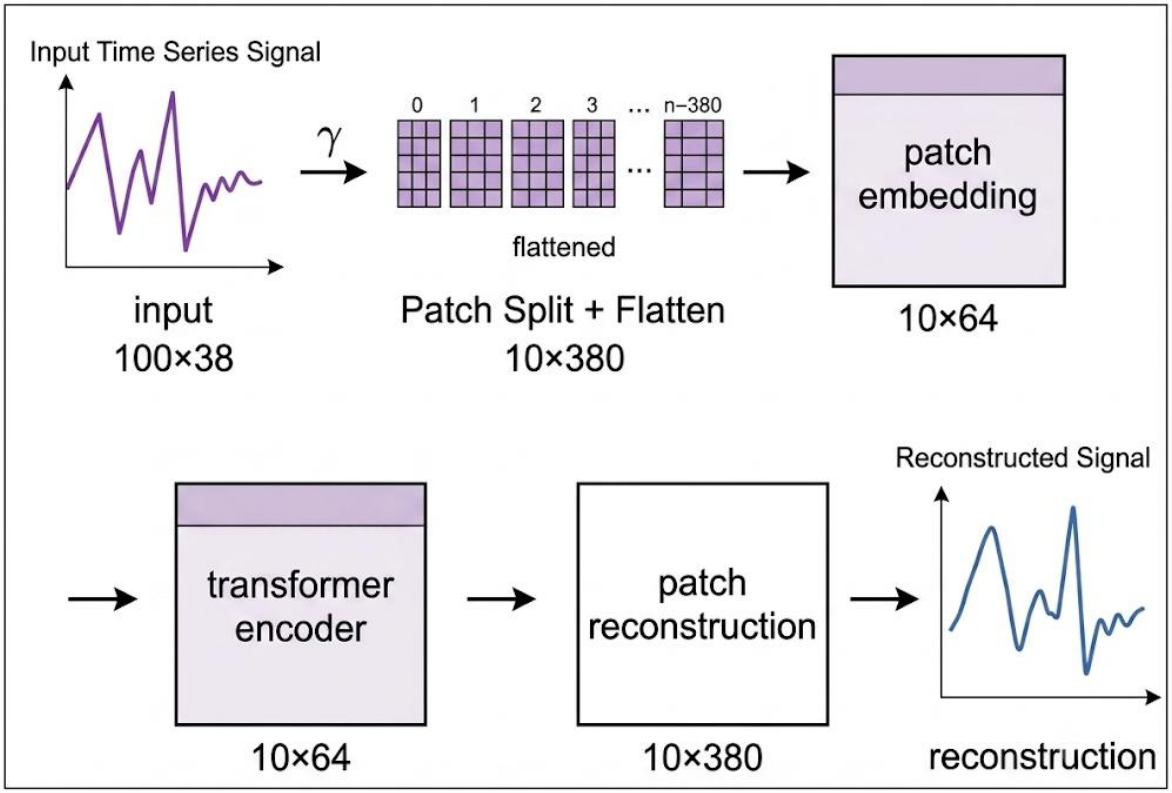


Figure 1. PWTA. Each window is split into patch tokens, encoded and reconstructed by a Transformer; the reconstruction error is the anomaly score.

Training minimizes the mean squared reconstruction error on normal windows. Since the training data is assumed to contain only normal behavior, the model is encouraged to learn regular temporal and cross-channel patterns. At inference time, the same reconstruction process is applied to test windows. The squared reconstruction error is averaged over channels to obtain a per-time-step error within each window. Because windows overlap, a

single time step may receive multiple error values; these values are merged by taking the maximum, producing one anomaly score for each time step. The overall procedure is summarized in Algorithm 1.

```
Algorithm 1  PWTA: training and inference
Input : train X_tr, test X_te, window W, patch P, stride S, epochs E
Output: per-step anomaly scores s


# Training
 1: W_tr <- MakeWindows(X_tr, W, S)
 2: init theta = {embed, posenc, transformer, recon}
 3: for epoch = 1..E:
 4:    for batch x in W_tr:
 5:       xp   <- Reshape(x, [n_patch, P*C])      # patch -> token
 6:       h    <- Transformer(embed(xp) + posenc)
 7:       xhat <- recon(h)
 8:       theta<- AdamStep(theta, grad ||xhat - x||^2)


# Inference
 9: W_te <- MakeWindows(X_te, W, S)
10: for window w_i in W_te:
11:    e_i <- mean_channel( (PWTA(w_i) - w_i)^2 )  # per-step error
12: s <- AggregateOverlap(e, max)                 # per-step score
13: return s
```

3.2 Difference from PatchTST

Although PWTA uses patch tokens, its purpose and design differ from PatchTST. PatchTST was developed for time-series forecasting and relies on channel independence: each channel is processed separately, and the model learns temporal patterns within each univariate series. PWTA, in contrast, is designed for anomaly detection through reconstruction. It deliberately combines all channels within each patch token, allowing reconstruction error to reflect not only unusual temporal patterns but also changes in cross-channel relationships.

Aspect	PWTA (ours)	PatchTST (2023)
Objective	anomaly detection (reconstruction)	forecasting
Channel handling	channels mixed within a token	channel independence

Table 1. PWTA and PatchTST differ in objective and channel handling.

This distinction is important because the patch representation itself is not the main contribution of PWTA. Rather, PWTA uses patches as a simple way to represent local temporal structure, while the reconstruction objective and channel-mixing design make it suitable for detecting deviations in multivariate time-series data.

3.3 Evaluation protocol

The methodological core of this work is the protocol, not the model. We remove the leniency of PA (Figure 2) and apply the following rules (Algorithm 2).

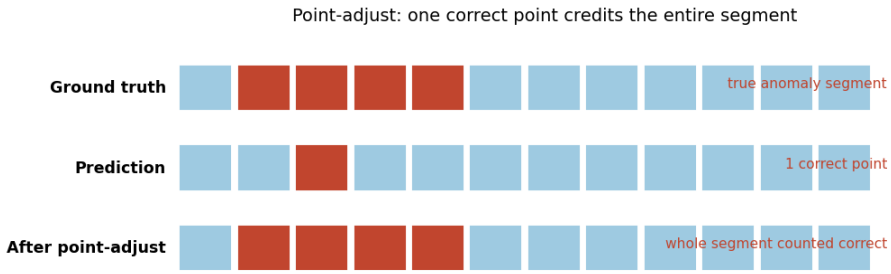


Figure 2. Point adjustment credits the whole segment after a single correct point, inflating F1.

1. raw PR-AUC. PR-AUC is computed directly from per-step scores and labels with no point adjustment. Because anomalies are rare, ROC is optimistic, so PR-AUC is the primary metric.
2. Random-normalized multiple. We report raw PR-AUC divided by the anomaly ratio; one means random, higher means real discrimination (Figure 6).
3. Per-entity decomposition. A pooled average is dominated by the dataset with the highest anomaly ratio, so we measure per machine or entity and report the distribution and the number of first-place wins.
4. PA F1 as a check. We also report point-adjusted F1 to confirm that our reproductions of the attention-based detectors match the published numbers, which validates the implementation.

```
Algorithm 2  Evaluation protocol (raw PR-AUC, no point adjustment)
Input : scores s, labels y, anomaly ratio r
1: prauc <- AveragePrecision(y, s)        # no point adjustment
2: mult  <- prauc / r                      # random-normalized
3: report prauc, mult per entity; then mean and #first-place
4: (check) paf1 <- F1(point_adjust(s>thr), y)    # must match paper
```

Chapter 4. Experimental Setting

4.1 Datasets

Dataset	Domain	Units	Channels	Anomaly ratio
SMD	server machines	28 machines	38	~0.042
PSM	server	1	25	0.278
MSL	Mars rover	27 entities	55	0.105
SMAP	satellite	55 entities	25	0.128
SWaT	water treatment (attack)	1	51	0.121

Table 2. Five real-world benchmarks. Training data is assumed normal.

We evaluate on five real-world TSAD benchmarks: SMD, SMAP, MSL, PSM, and SWaT. SMD, SMAP, and MSL are multi-entity datasets, containing 28 server machines, 55 satellite entities, and 27 Mars rover entities, respectively. These three datasets form the 110-entity group used in the main per-entity comparison. PSM and SWaT are treated as single dataset-level benchmarks, so they are reported separately rather than included in the 110-entity count. Training data is assumed to contain only normal behavior, and labels are used only for evaluation. For SMAP and MSL, segment annotations are converted into point-level binary labels. PSM and SWaT are normalized using train-based z-score normalization because their raw scales are large; SMD, SMAP, and MSL are left unnormalized.

4.2 Model, training, and tuning

Hyperparameter	Value
window / stride / patch	100 / 10 / 10 (n_patch=10)
d_model / heads / layers	64 / 4 / 2
dropout	0.1
epochs / batch / lr	50 / 128 / 1e-3
optimizer / loss	Adam / MSE

Table 3. PWTA hyperparameters, fixed across all datasets.

Experiments run on two NVIDIA L40S GPUs with PyTorch 2.5.1 (CUDA 12.4). We fix the seed to 0 and verify stability across seeds 0/1/2 for the reproduced attention-based detectors comparison. Window and patch sizes are held constant across datasets so that no dataset receives bespoke tuning; robustness to patch size is examined in Section 5.5.

4.3 Baselines, AnomalyTransformer, and DCdetector

Classical baselines are PCA reconstruction, Isolation Forest, 1-NN distance, LOF, and random scoring. We also evaluate two recent attention-based detectors, AnomalyTransformer and DCdetector, using their official implementations with a minimal change to save the score arrays. For AnomalyTransformer, we set input_c and output_c to the channel count and use the official three training epochs, since the learning curve plateaus by epoch 3.

Chapter 5. Results and Discussion

5.1 Point-adjusted F1 is misleading

We first verify that the reproduced implementation is faithful to the published setting. With the official configuration, AnomalyTransformer reproduces the reported PA-based F1 scores closely on MSL and SMAP (Table 4). However, when the same score outputs are evaluated with raw PR-AUC, performance is close to the random baseline (Figure 3).

Dataset	Paper F1	Repro. F1	Match	raw PR-AUC	vs random
MSL	0.916	0.935	Yes	0.105	1.00x
SMAP	0.967	0.965	Yes	0.129	1.01x

Table 4. AnomalyTransformer reproduction. PA F1 matches the paper; raw PR-AUC is random.

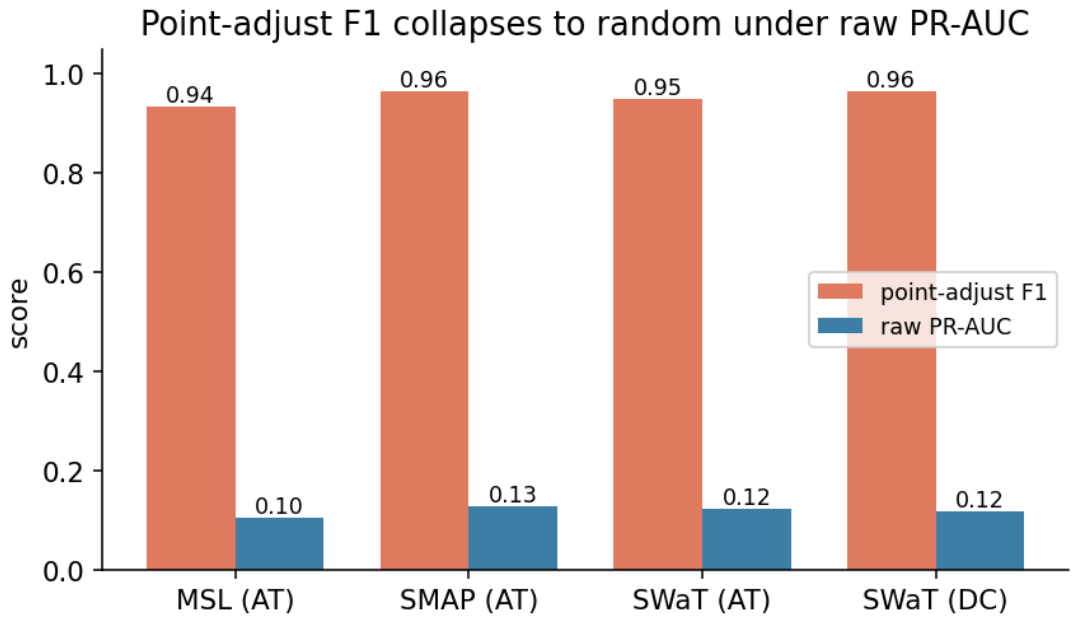


Figure 3. PA F1 (orange) reaches 0.95 while raw PR-AUC (blue) equals random.

This result suggests that the low raw PR-AUC is not caused by a failed reproduction. The

model can recover the published PA F1 scores, but those scores do not correspond to meaningful point-level discrimination. The underlying mechanism is visible in the score distribution on SWaT (Table 5, Figure 4). A large fraction of scores are exactly zero, ranging from 86.5% for AnomalyTransformer to 97.9% for DCdetector. Moreover, the zero fraction is slightly higher inside anomaly regions than in normal regions, indicating weak or reversed separation. Even after excluding zero scores, the ROC-AUC remains close to random. These results show how PA can produce high F1 values despite limited raw discriminative signal.

Diagnostic	AnomalyTransformer	DCdetector
zeros (overall)	86.5%	97.9%
zeros in anomaly regions	92.9%	98.0%
zeros in normal regions	85.5%	97.9%
ROC-AUC, zeros excluded	0.522	0.515
raw PR-AUC / PA F1	0.124 / 0.949	0.119 / 0.964

Table 5. Score degeneracy of the two models on SWaT.

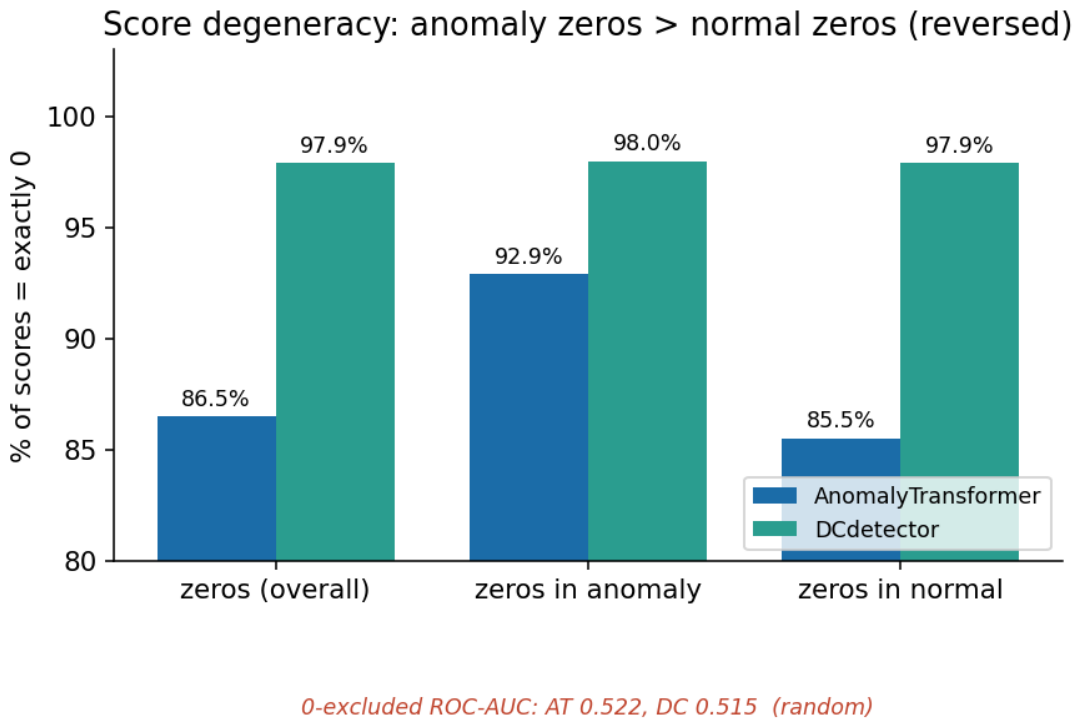


Figure 4. Anomaly regions have more zeros than normal regions; the non-zero ROC is random.

5.2 Per-entity comparison on SMD, SMAP, and MSL

Under the same metric, PWTA ranks first across the 110 entities (SMD 28 + SMAP 55 + MSL 27): PWTA 0.431, 1-NN 0.317, PCA 0.290, LOF 0.267, and Isolation Forest 0.212 (Figure 5).

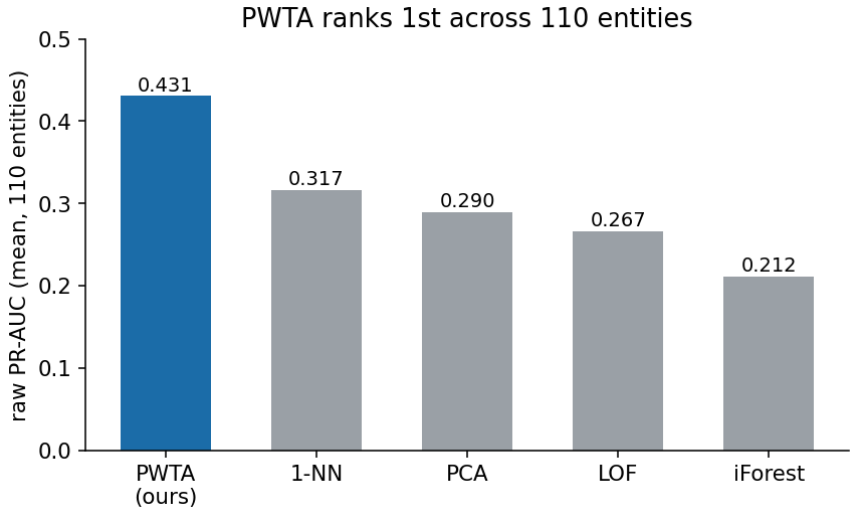


Figure 5. Mean raw PR-AUC over 110 entities; PWTA leads at 0.431.

Because the anomaly ratio differs across datasets, we also report the random-normalized multiple, defined as raw PR-AUC divided by the anomaly ratio. PWTA again gives the largest factor over random (Figure 6).

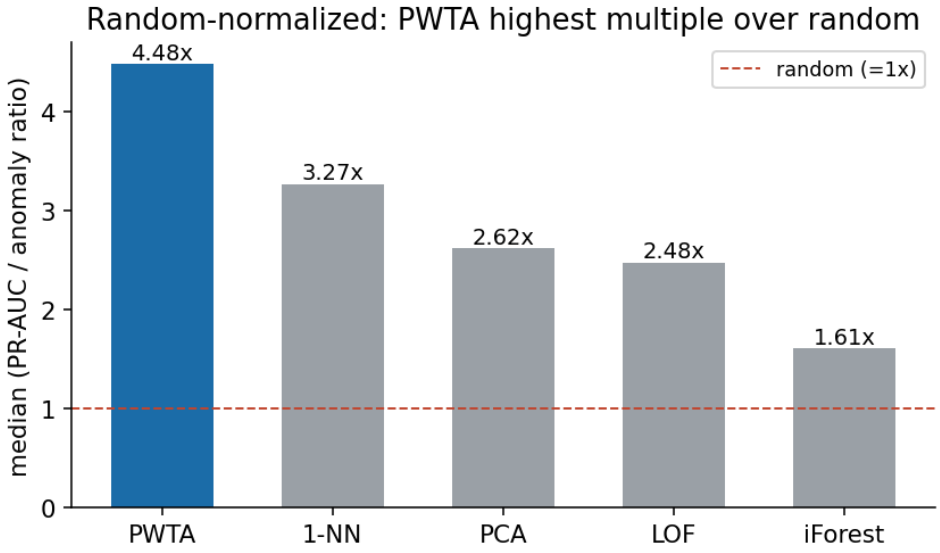


Figure 6. Median PR-AUC divided by anomaly ratio. PWTA has the largest multiple over random.

PWTA also ranks first on 51 of the 110 entities, ahead of the runner-up 1-NN, which ranks first on 26 entities (Figure 7). Compared entity by entity against the runner-up, PWTA lies above the diagonal for most entities (Figure 8).

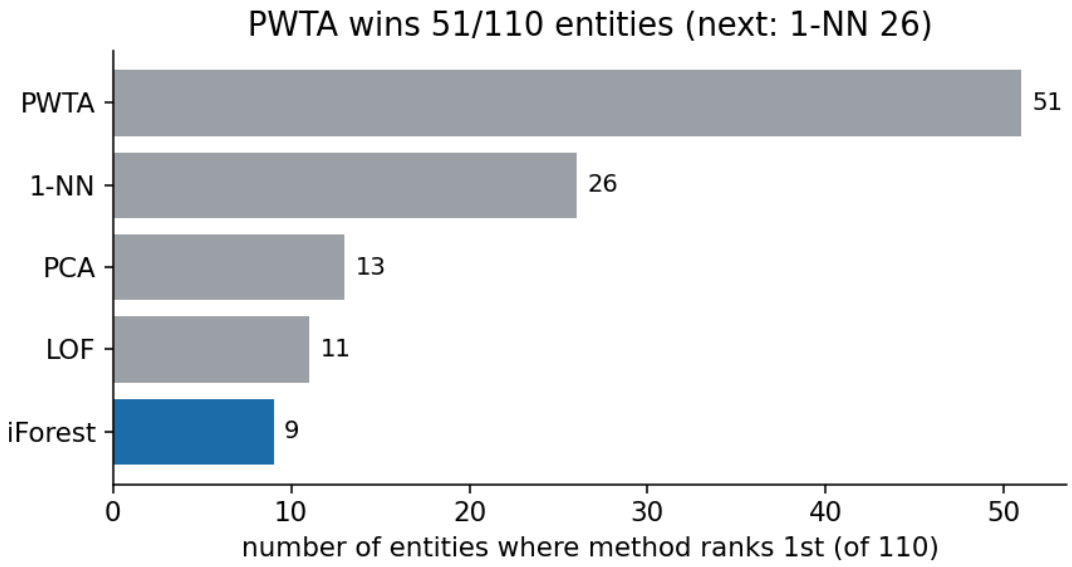


Figure 7. First-place counts. PWTA wins 51 of 110 entities; 1-NN is next at 26.

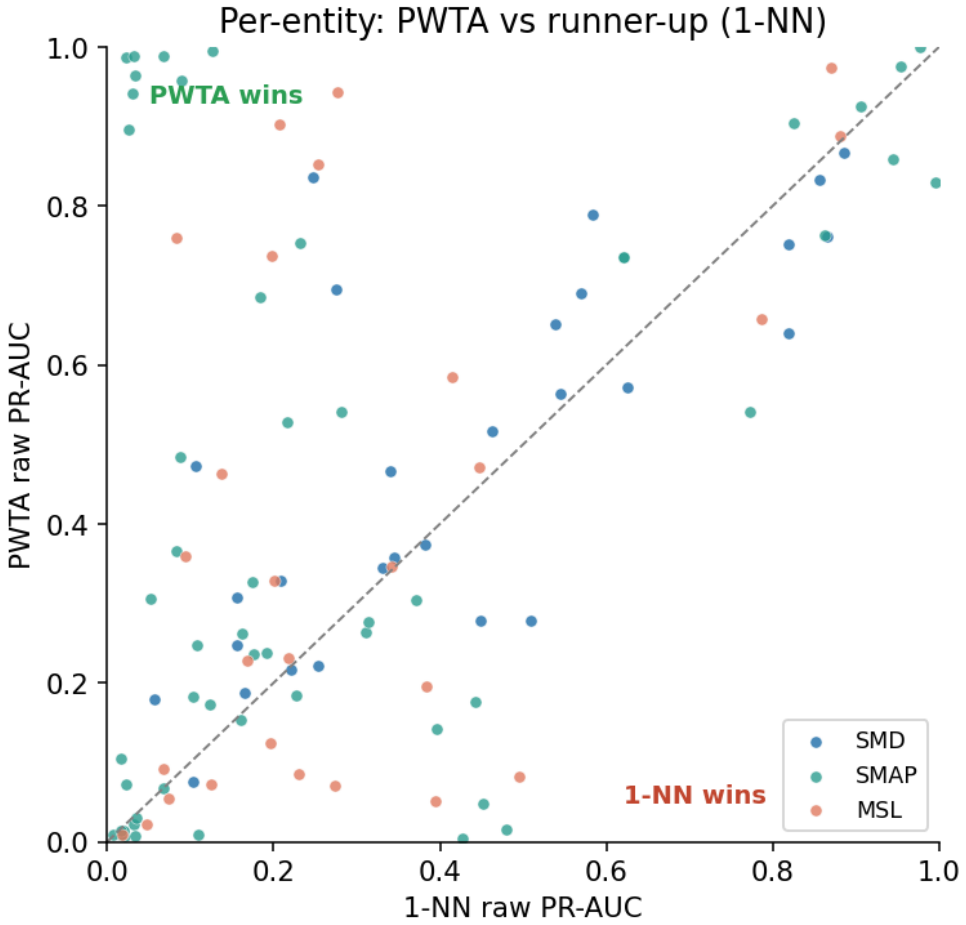


Figure 8. Per-entity PWTA vs runner-up 1-NN. Points above the diagonal indicate entities where PWTA performs better.

The lead also holds within the three multi-entity benchmarks: SMD 0.482, SMAP 0.423, and MSL 0.392. PWTA ranks first on 14 of 28 SMD machines, 25 of 55 SMAP entities, and 12 of 27 MSL entities. PSM is reported separately because it is treated as a single dataset-level benchmark; on PSM, PWTA obtains a raw PR-AUC of 0.556 and outperforms all four classical baselines.

5.3 Comparison with AnomalyTransformer under raw PR-AUC

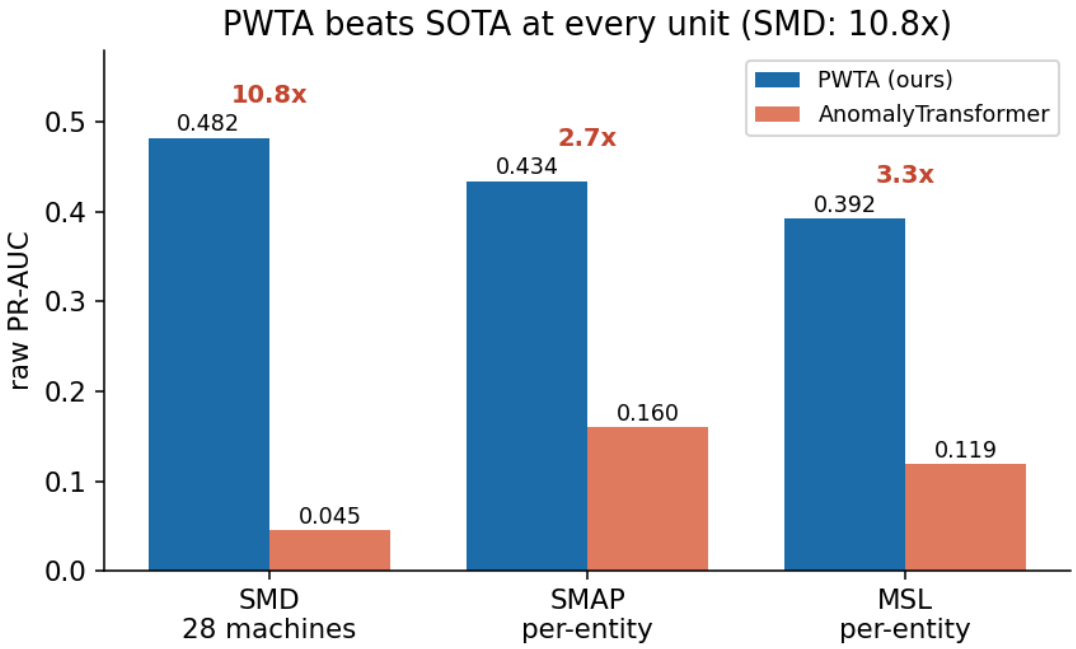


Figure 9. PWTA vs AnomalyTransformer under raw PR-AUC

We next compare PWTA with AnomalyTransformer under the same raw point-level metric. This comparison is important because AnomalyTransformer reproduces its reported point-adjusted F1 scores, but its raw PR-AUC remains close to the anomaly ratio. Therefore, the comparison tests whether high point-adjusted F1 corresponds to meaningful point-level anomaly-score discrimination. PWTA achieves substantially higher raw PR-AUC than AnomalyTransformer on all three multi-entity benchmarks. The largest difference appears on SMD, where PWTA obtains a raw PR-AUC of 0.482, compared with 0.045 for AnomalyTransformer. This corresponds to a 10.8-fold improvement. The gains remain substantial on SMAP and MSL, with multiples of 2.7 and 3.3, respectively.

These results should not be interpreted as a claim that PWTA dominates all recent time-series anomaly detection models. Rather, they show that under raw point-level PR-AUC, PWTA provides stronger anomaly-score discrimination than AnomalyTransformer in the evaluated multi-entity benchmarks.

5.4 Where the advantage holds, and why

PWTA does not perform best in every setting. Its main failure case is SWaT, where Isolation Forest achieves substantially higher raw PR-AUC than PWTA. This difference suggests that the two methods respond to different forms of abnormality. To examine this boundary, we analyze 327 anomaly segments from SMD and compare point-level recall with raw segment properties such as deviation magnitude and duration.

The results show that deviation magnitude is more strongly related to recall than segment duration. Magnitude correlates with recall at +0.469 ($p = 2.5e-19$), whereas duration has a weak and statistically insignificant correlation of -0.062 ($p = 0.26$). After controlling for magnitude, duration remains close to zero (Figure 10, Figure 11). This indicates that missed anomalies are not primarily long anomalies, but low-magnitude anomalies.

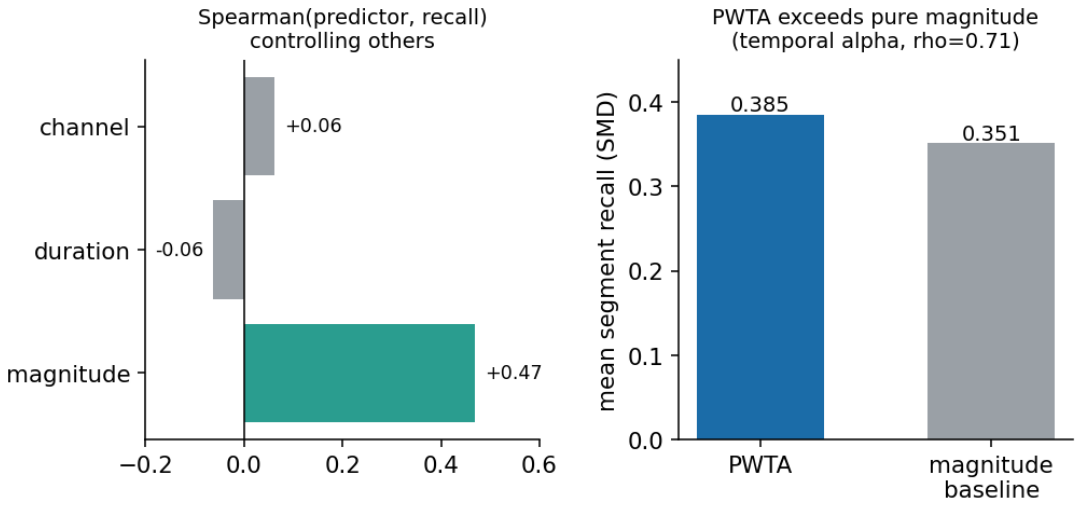


Figure 10. (left) Magnitude predicts recall, duration does not. (right) PWTA exceeds a pure magnitude baseline.

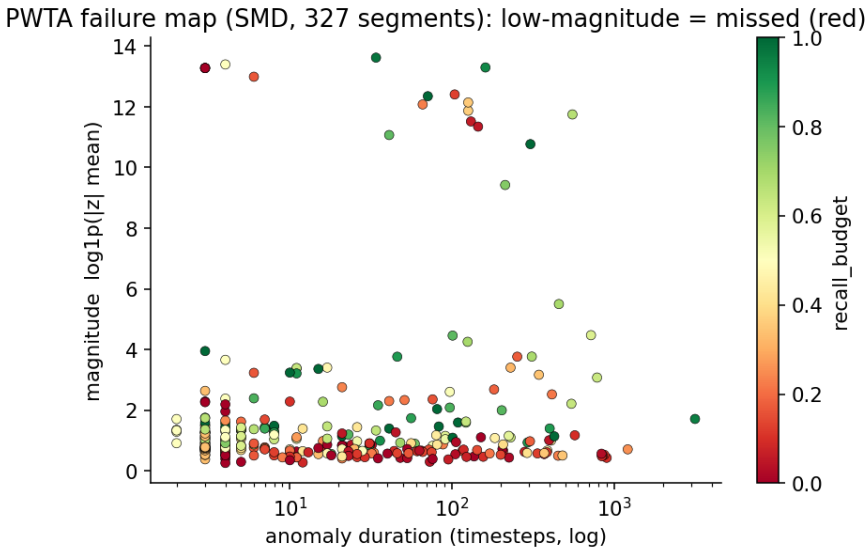


Figure 11. Failure map over 327 SMD segments. The low-magnitude band is red (missed), independent of duration.

We also test whether PWTA is merely acting as a magnitude detector. The answer appears to be no. PWTA achieves higher recall than a learning-free magnitude baseline, and some anomaly segments are detected by PWTA but missed by the magnitude baseline (Figure 12). This suggests that PWTA captures temporal information beyond simple deviation size, although the margin is still modest.

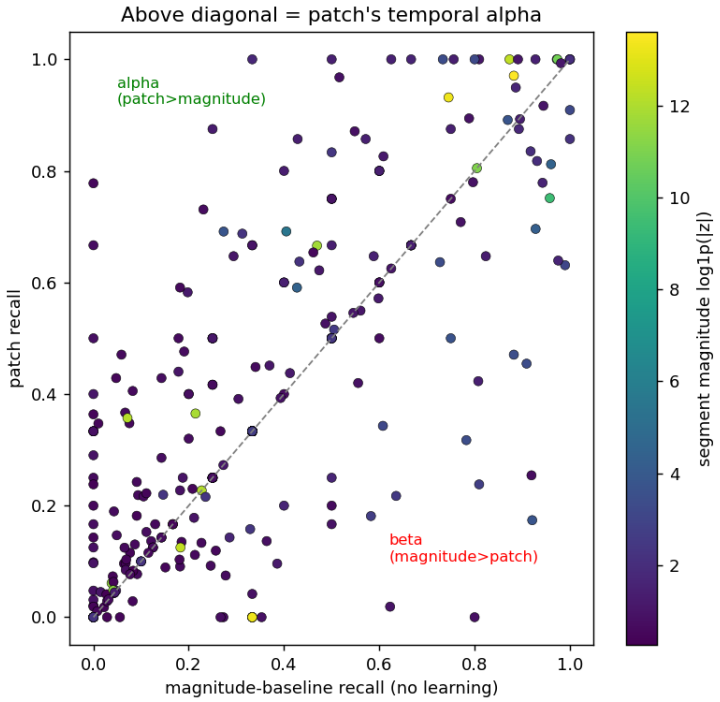


Figure 12. PWTA recall vs magnitude-baseline recall. Points above the diagonal are PWTA's temporal gain.

SWaT falls into a different regime. Many of its anomalies remain within the normal value range, and the median deviation magnitude is lower than in SMD. In this setting, reconstruction error is less effective because the anomalous values are not necessarily far from the training distribution in magnitude. Isolation Forest, by contrast, can separate points according to their position in the multivariate value distribution, which explains its stronger performance on SWaT (Figure 13). Thus, the limitation of PWTA is not only a performance failure, but also evidence of an inductive-bias boundary: reconstruction-based detection is better suited to time-pattern anomalies, whereas isolation-based detection can be more effective for value-range anomalies.

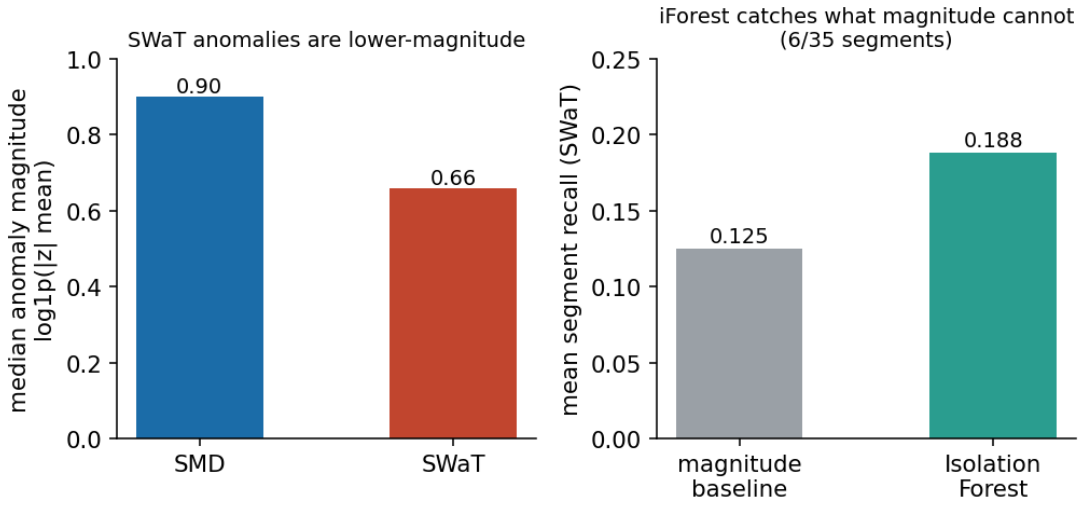


Figure 13. SWaT anomalies are lower-magnitude (left), and isolation catches what magnitude cannot (right).

5.5 Robustness to patch size

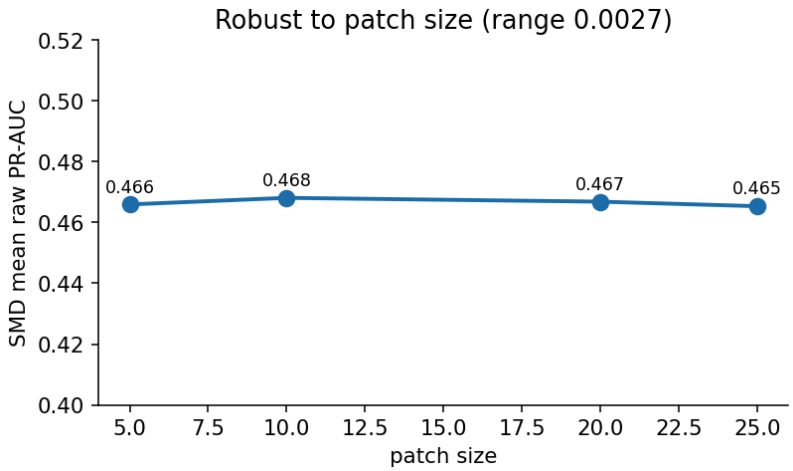


Figure 14. Mean SMD raw PR-AUC across patch sizes 5/10/20/25 varies by 0.0027; the advantage is not a tuning artifact.

Chapter 6. Conclusion

Removing point adjustment and evaluating models with raw PR-AUC across 110 entities leads to a substantially different view of TSAD performance. Models that reproduce their published PA-based F1 scores perform close to random under the raw metric, and our analysis shows that this behavior is associated with degenerate score distributions, including a large mass of exact zeros and weak or reversed separation between normal and anomalous regions. In contrast, PWTA, a simple patch-wise reconstruction model, outperforms five classical baselines and AnomalyTransformer in the per-entity comparison, ranking first on 51 of the 110 entities and exceeding AnomalyTransformer by 10.8 times on SMD.

The results also clarify the conditions under which this advantage holds. PWTA performs well on datasets where anomalies are expressed mainly through temporal or cross-channel pattern changes. However, its advantage does not extend to all anomaly types. On SWaT, where many anomalies remain within the normal value range, Isolation Forest performs substantially better. This contrast suggests that reconstruction error and isolation-based scoring capture different forms of abnormality. PWTA is effective when anomalous behavior disrupts learned temporal structure, whereas isolation-based methods are better suited to anomalies that are separated by their position in the value distribution.

Overall, the findings emphasize the importance of evaluation protocol before architectural complexity. Under PA-based F1, weak point-level discrimination can appear strong; under raw PR-AUC, the same models may provide little useful separation. Once this evaluation artifact is removed, a simple reconstruction-based method can outperform more elaborate Transformer-based detectors in several time-pattern anomaly regimes. This suggests that part of the reported progress in TSAD should be revisited under raw, point-level metrics.

Several limitations remain. First, the AnomalyTransformer and DCdetector comparison in this work is limited to two attention-based models. Extending the study to structurally different models, such as TranAD, would help determine whether the observed weakness is specific to attention-based detectors or reflects a broader evaluation issue. Second, although PWTA outperforms the compared methods, its absolute performance is still modest in some domains, reaching only around 1.3 times the random baseline. Thus, PWTA should be viewed as stronger than the evaluated alternatives under this protocol, but not as a solved detector. Finally, the temporal advantage over a magnitude-based baseline should be tested more broadly across datasets, and future work should reduce over-generalization in reconstruction to improve absolute detection performance. Additional rejected hypotheses and diagnostic analyses are provided in the Appendix.

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Appendix. Rejected hypotheses and diagnostics

The main text presents only the line of argument that survived. For honesty we record here the hypotheses tried and discarded.

1. Duration hypothesis (rejected). The early idea that long anomalies are missed failed under partial correlation (duration | magnitude rho = -0.057, p = 0.30). The signal seen in single machines and pooled PR-AUC was a shadow of magnitude and of machine-level confounding.
2. Sample entropy (rejected). Predicting reconstruction failure from the sample entropy of normal segments failed (rho = -0.28, p = 0.38).
3. Lag-1 ACF (rejected). Did not separate improved from unimproved machines.
4. Improvement attempts. Score normalization (ch_norm_mean) gave +3 percent and is worth adopting; a memory bank was a wash; channel attention regressed, because the base model already mixes channels.
5. Circularity check. Recall comes from reconstruction error, which can be tied to magnitude by construction; we control for this by testing whether PWTA exceeds a magnitude baseline (Section 5.4, diagnostic 2).